

Design of Protograph-Based LDPC Codes for Rate-Diverse Multiple Access

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Abstract—In this paper, we investigate the performance of protograph-based low-density parity-check (LDPC) codes for rate-diverse two-user Gaussian multiple access channel (GMAC). Considering the characteristic of rate-diverse GMAC, we first analyze the message passing model of the coding system. Then, we derive a rate-diverse joint protograph-based extrinsic information transfer (RDJP-EXIT) chart to analyze the rate-diverse joint user messages decoding (RDJD) behavior. Guided by the RDJP-EXIT, we design the protograph base matrices of different channel conditions for two users, respectively. Simulation results show that the proposed protograph-based LDPC codes of low decoding threshold significantly outperform the conventional AR4JA codes and irregular LDPC codes in GMAC channels, which are also close to optimal in terms of sum rate.

Index Terms—Two-user GMAC, LDPC codes, protograph codes, joint decoding.

I. INTRODUCTION

Gaussian Multiple Access Channel (GMAC) is a multiuser channel that has received much attention in the 5G and future communication technologies [1]. The capacity region of the two-user GMAC was characterized in [1] and [2], where the corner points of the region can be achieved by single user decoding, and other points can be achieved by time-sharing, rate-splitting, or joint decoding. Channel coding can enhance the performance of GMAC [3]. [4] and [5] used extrinsic information transfer (EXIT) chart analysis to optimize irregular low-density parity-check (LDPC) codes for two-user GMAC. More recently, protograph-based LDPC codes were designed for GMAC due to superior performance and low complexity of structured coding [6].

Physical layer network coding (PNC) improves GMAC performance by converting received superimposed signals into useful network coded (NC) messages [7]. Network-coded multiple-access (NCMA) employs XOR-based channel decoding (XOR-CD) and multiuser decoding (MUD) to decode the NC message and then the two user messages [8]. The decoding for compute-and-forward multiple-access (CFMA) is similar to that of NCMA [9]. The achievable rate of XOR-CD for both schemes is limited by the poor user channel. Thus, [10] developed a set of encoding and rate-diverse joint user messages decoding (RDJD). It

performs better than the NCMA and CFMA, especially in significantly different user channels. By iterating decoding within a parity-check matrix, it also outperforms the traditional SIC even in the near power-balanced GMAC. However, to best of our knowledge, there is still no research on the coding design for the RDJD.

Thus, this paper proposes the design of protograph-based LDPC codes for rate-diverse GMAC. We first derive a rate-diverse joint protograph-based extrinsic information transfer (RDJP-EXIT) chart to characterize the RDJD behavior. Then, we design the protograph matrices for two users guided by the RDJP-EXIT. Simulation results show that the proposed protograph LDPC codes significantly outperform the AR4JA codes and irregular codes over GMAC with various channels.

II. BACKGROUND

In this paper, vectors are represented by bold symbols, i.e., $\mathbf{x}$, and their elements are indicated by the italic ones, i.e., x .

A. Two-User GMAC Model

The GMAC model consists of two users that need to simultaneously transmit information to a common receiver. The source message vector $\mathbf{u}^{[k]}$ of length F_k for user k is encoded into $\mathbf{v}^{[k]}$ of length L , $k = 1, 2$. Let $\mathbf{x}^{[k]}$ be the BPSK modulated codewords of $\mathbf{v}^{[k]}$. At the receiver, the output signal of the GMAC channel is given by

$$\mathbf{y} = \sqrt{P_1}\mathbf{x}^{[1]} + \sqrt{P_2}\mathbf{x}^{[2]} + \mathbf{z}, \quad (1)$$

where $P_k = E[h_k^2]$ denotes the power gains, which are the expectation of channel coefficients h_k for user k , and $\mathbf{z}$ contains additive white Gaussian noise with zero mean and variance σ^2 .

B. Rate-Diverse Joint Decoding for GMAC

In the rate-diverse GMAC, channel codes are assigned to users based on their channel quality, with a high code rate for the better channel and a low code rate for the inferior one [10]. We assume that strong user 1 adopts a code $\mathbf{v}^{[1]}$ of rate R_1 with a parity-check matrix $\mathbf{H}_1$, while weak user 2 adopts a code $\mathbf{v}^{[2]}$ of rate R_2 with a parity-check matrix $\mathbf{H}_2$. As given in [10], we have the format of the matrices $\mathbf{H}_1$ and $\mathbf{H}_2$ as

$$\mathbf{H}_2 = \begin{pmatrix} \mathbf{H}_1 \\ \mathbf{H}_e \end{pmatrix}. \quad (2)$$

We see that $\mathbf{H}_2$ is composed by adding a residual matrix $\mathbf{H}_e$ to the end of $\mathbf{H}_1$. Let $\mathbf{H}_c$ denote the common matrix of $\mathbf{H}_1$ and $\mathbf{H}_2$, i.e., $\mathbf{H}_c = \mathbf{H}_1$. We define matrix $\mathbf{v}^{[1,2]} = [\mathbf{v}^{[1]}; \mathbf{v}^{[2]}]$ as the joint codeword of $\mathbf{v}^{[1]}$ and $\mathbf{v}^{[2]}$, where the j -th element contains two coded bits $v_j^{[1,2]} = [v_j^{[1]}, v_j^{[2]}]$, $v_j^{[1,2]} \in \{0, 1\} \times \{0, 1\}$, $j = 1, 2, \dots, L$. Under this structure, we have

$$\begin{aligned} \mathbf{v}^{[1,2]}\mathbf{H}_c^T &= 0 \quad \text{for users 1 and 2,} \\ \mathbf{v}^{[2]}\mathbf{H}_e^T &= 0 \quad \text{for user 2.} \end{aligned} \quad (3)$$

Therefore, in RDJD, we use a common joint user decoder (JUD) within $\mathbf{H}_c$ to decode joint codeword $\mathbf{v}^{[1,2]}$, while a residual user decoder (RUD) associated with $\mathbf{H}_e$ decodes $\mathbf{v}^{[2]}$. As shown in Fig. 1, the JUD first uses channel information $\mathbf{y}$ and the posterior information of $\mathbf{v}^{[2]}$ from RUD to decode $(\mathbf{v}^{[1]}, \mathbf{v}^{[2]})$, and then the RUD uses the marginal information of $\mathbf{v}^{[2]}$ from $(\hat{\mathbf{v}}^{[1]}, \hat{\mathbf{v}}^{[2]})$ to estimate $\mathbf{v}^{[2]}$. This process continues iteratively until the decoding results satisfy $\hat{\mathbf{v}}^{[1]}\mathbf{H}_1^T = 0$ and $\hat{\mathbf{v}}^{[2]}\mathbf{H}_2^T = 0$, or the maximum number of iterations is reached.

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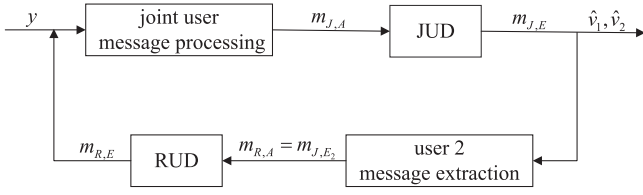


Fig. 1. Decoding diagram of RDJD.

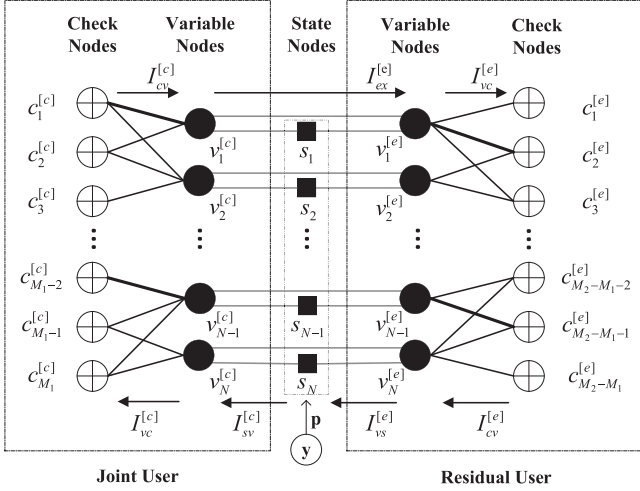


Fig. 2. A joint protograph for JU and RU.

In the protograph $\mathbf{G} = (\mathbf{V}, \mathbf{C}, \mathbf{E})$, each edge $e_{i,j} \in \mathbf{E}$ connects a variable node $v_j \in \mathbf{V}$, $j = 1, 2, \dots, N$ with a check node $c_i \in \mathbf{C}$, $i = 1, 2, \dots, M$. By copying and permuting operation for the protograph, a LDPC code of desired length is obtained. Furthermore, the base matrix $\mathbf{B}$ associated with the protograph and can be analyzed by the EXIT for protograph codes (PEXIT) [11], [12].

III. PROPOSED JOINT PROTOGRAPH ANALYSIS

A. Proposed Message Model Analysis

In the rate-diverse GMAC, $\mathbf{H}_e$ is a matrix added to $\mathbf{H}_c$, as shown in (2). Let $\mathbf{B}^{[c]} = (b_{i,j})_{M_1 \times N}$ and $\mathbf{B}^{[e]} = (b_{i,j})_{(M_2-M_1) \times N}$ denote the base matrices of $\mathbf{H}_c$ and $\mathbf{H}_e$, respectively, where M_k is the number of check nodes for user k . Fig. 2 shows the joint protograph for the joint user (JU) and the residual user (RU). The bold lines represent the parallel edges that are permissible within the protograph.

1) *JU*: According to (3), the channel decoder for $\mathbf{H}_c$ is capable of decoding $\mathbf{v}^{[1,2]}$. Let ρ be the conversion function from binary vector to decimal number, i.e., $\theta = \rho(v_j^{[1,2]}) = \rho([v_j^{[1]}, v_j^{[2]}])$, where $\theta \in \Theta = \{0, 1, 2, 3\}$. Let $\rho_m^{-1}(\theta)$ be the m -th bit of the binary vector obtained from θ , $m = 1, 2$. Decoding the joint messages is similar to that on $\mathbb{F}_{2^2}$ [13], thus we employ the non-binary PEXIT analysis for $\mathbf{H}_c$.

To achieve the symmetry of message passing in JU, we employ the scheme proposed in [14]. Let the random sequences be denoted as $\mathbf{t}_1 = \{t_1(1), t_1(2), \dots, t_1(N)\}$ and $\mathbf{t}_2 = \{t_2(1), t_2(2), \dots, t_2(N)\}$, where each element is equally likely to take the values of 1 or -1. The users transmit $\mathbf{t}_1 \mathbf{x}^{[1]}$ and $\mathbf{t}_2 \mathbf{x}^{[2]}$ respectively. This method can achieve channel symmetry and simplifies the decoding threshold estimation by hypothesizing that all transmitted codewords are zero.

• *Initial messages for JU*: Let $x_j = \sqrt{P_1} t_1(j) x_j^{[1]} + \sqrt{P_2} t_2(j) x_j^{[2]}$ denote a superimposed symbol with $\sqrt{P_k}$, $x_j \in \mathcal{X} = \{\sqrt{P_1}, \sqrt{P_2}, \sqrt{P_1} - \sqrt{P_2}, -\sqrt{P_1} + \sqrt{P_2}, -\sqrt{P_1} - \sqrt{P_2}\}$ and $\mathcal{X}^i$ is the i -th

element in $\mathcal{X}$, $j = 1, \dots, N$. Fig. 2 shows that at the state node s_j , we use a 4-dimensional probability vector $\mathbf{p}_j$ for $\mathbf{y}$, $\mathbf{p}_j = [p_{00}, p_{01}, p_{10}, p_{11}]$. Let $\mathbf{p}_j^i$ be the i -th element of the $\mathbf{p}_j$, expressed as

$$\begin{aligned} \mathbf{p}_j^i &= \Pr\{\mathbf{y}_j | x_j = \mathcal{X}^i, \sqrt{P_1}, \sqrt{P_2}\} \\ &= \alpha_j \cdot \exp\left(-\frac{(\mathbf{y}_j - \mathcal{X}^i)^2}{2\sigma^2}\right), \end{aligned} \quad (4)$$

where α_j is a normalization term satisfying $\sum_{i=0}^3 \mathbf{p}_j^i = 1$.

All messages in the graph are transmitted in the form of log-likelihood ratio (LLR) among different nodes. Let $m_{vs}^{[e]}$ denote the message from the variable node to the state node for RU, and let $m_{sv}^{[c]}$ denote the state-to-variable message for JU. The initial message of JU can be computed by

$$m_{sv}^{[c]} = \log \frac{e^{-\frac{(y - \sqrt{P_1} - \sqrt{P_2})^2}{2\sigma^2}} e^{m_{vs}^{[e]}} + e^{-\frac{(y + \sqrt{P_1} + \sqrt{P_2})^2}{2\sigma^2}}}{e^{-\frac{(y + \sqrt{P_1} - \sqrt{P_2})^2}{2\sigma^2}} e^{m_{vs}^{[e]}} + e^{-\frac{(y - \sqrt{P_1} + \sqrt{P_2})^2}{2\sigma^2}}}. \quad (5)$$

Following the decoding process of the RDJD, the state node s is introduced only in the messages passed from RU to JU. Recall that both users transmit all-zero codewords in the symmetric channels, leading to a probability of $\frac{1}{4}$ for each received value of $x \in \mathcal{X}$. In particular, we derive the functions $F_{+\sqrt{P_1}+\sqrt{P_2}}, F_{+\sqrt{P_1}-\sqrt{P_2}}, F_{-\sqrt{P_1}+\sqrt{P_2}}$ and $F_{-\sqrt{P_1}-\sqrt{P_2}}$ to represent the mean value of $m_{sv}^{[c]}$ in (5) for the four cases in (6)–(9) shown at the bottom of next page, respectively. The $m_{vs}^{[e]}$ from RU is distributed as $N(\mu, 2\mu)$ and σ_j^2 is the noise variance of s_j .

• *Message update between variable nodes and check nodes*: For non-binary LDPC code over $\text{GF}(q)$, under the assumptions of uniform random edge weights and coset vectors, message symmetry and permutation-invariance were demonstrated [15]. Supported by this, we have following message distributions.

Check-to-variable message distribution: The distribution of check-to-variable (c-to-v) messages can be modeled as a Gaussian random vector of dimension $(q-1)$ with mean $\mathbf{m}$ and covariance matrix Σ as

$$\mathbf{m} = \begin{bmatrix} \sigma^2/2 \\ \sigma^2/2 \\ \vdots \\ \sigma^2/2 \end{bmatrix}_{(q-1) \times 1}, \quad \Sigma = \begin{bmatrix} \sigma^2 & & & \sigma^2/2 \\ & \sigma^2 & & \\ & & \ddots & \\ \sigma^2/2 & & & \sigma^2 \end{bmatrix}. \quad (10)$$

Variable-to-check message distribution: The distribution of variable-to-check (v-to-c) messages are conceived as the sum of an initial message from the channel and c-to-v messages.

Furthermore, the mutual information (MI) I between the transmitted symbol S and the random LLR-vector $\mathbf{W}$ of the transmitted messages can be computed as

$$I(S; \mathbf{W}) = 1 - \mathbb{E} \left[\log_q \left(1 + \sum_{i=1}^{q-1} e^{-W_i} \right) | S = 0 \right], \quad (11)$$

where $W_i = \log(P_0/P_i)$ and P_i represents the posterior probability distribution of $i = \theta \in \Theta$, $\sum_{i=0}^3 P_i = 1$. The MI function between c-to-v messages and transmitted symbols is denoted by $J_A(\sigma)$, where σ is a parameter satisfying (10). The v-to-c MI function can be denoted as $J_R(\sigma_c, \sigma)$, where σ_c^2 is the initial message variance. Since the non-binary MI functions do not have a closed-form expression, the Monte Carlo method is utilized to approximate these functions [16].

2) *RU*: Before RU decoding, we extract the *a priori* information of $\mathbf{v}^{[2]}$ from LLR-vector $\mathbf{W}$. Let $\mathbf{a} = [a_0, a_1]$, and a_n denotes the

probability of $v_j^{[2]} = n$. Given $\mathbf{W}$, the probability vector component can be given by

$$P_i = \text{LLR}_i^{-1}(\mathbf{W}) = \frac{e^{-W_i}}{1 + \sum_{k=1}^3 e^{-W_k}}, \quad i = 0, \dots, 3. \quad (12)$$

Then, we extract the probability message for user 2 from the posterior probability of JU, calculated by

$$a_n = \sum_{\rho_2^{-1}(\theta)=n, \theta \in \Theta} P_\theta. \quad (13)$$

Thus, we obtain the prior-LLR $U = \log(a_0/a_1)$ for RU.

Additionally, under the Gaussian Approximation, the binary MI passed between v-and-c is given by

$$I(X; G) = J(\sigma_m) = 1 - \int_{-\infty}^{\infty} \frac{e^{-\frac{(\xi - \sigma_m^2/2)^2}{2\sigma_m^2}}}{\sqrt{2\pi\sigma_m^2}} \log_2(1 + e^{-\xi}) d\xi, \quad (14)$$

which can be efficiently calculated by piecewise expression approximation [17]. Here, X denotes a binary coded bit and its corresponding LLR-value $G \sim N(\sigma_m^2/2, \sigma_m^2)$.

B. Proposed RDJP-EXIT Analysis

Fig. 2 illustrates the RDJP-EXIT analysis, where the MI between nodes of JU and RU is updated along the edges of this joint protograph. We define $I_{ab}^{[c]}$ ($I_{ab}^{[e]}$) to denote the MI from node a to node b for JU (RU), where (a, b) can refer to variable nodes v , check nodes c , or state check nodes s . The detail analysis of the RDJP-EXIT is presented as follows, noting that $i = 1, \dots, M$ and $j = 1, \dots, N$.

• *V-to-s message update for RU*: Assuming iteration starts from JU, the message passed from RU to JU is given by

$$I_{vs}^{[e]}(j) = J \left(\sqrt{\sum_{i=1}^M b_{ij} [J^{-1}(I_{cv}^{[e]}(i, j))]^2} \right). \quad (15)$$

• *S-to-v message update for JU*: The initial message received by JU from the state node is denoted as

$$I_{sv}^{[c]}(j) = \frac{1}{4} J \left(\sqrt{2 \left| F_{+\sqrt{P_1}, +\sqrt{P_2}} \left(\frac{1}{2} [J^{-1}(I_{vs}^{[e]}(j))]^2, \sigma_j^2 \right) \right|} \right)$$

$$+ \frac{1}{4} J \left(\sqrt{2 \left| F_{+\sqrt{P_1}, -\sqrt{P_2}} \left(\frac{1}{2} [J^{-1}(I_{vs}^{[e]}(j))]^2, \sigma_j^2 \right) \right|} \right) \\ + \frac{1}{4} J \left(\sqrt{2 \left| F_{-\sqrt{P_1}, +\sqrt{P_2}} \left(\frac{1}{2} [J^{-1}(I_{vs}^{[e]}(j))]^2, \sigma_j^2 \right) \right|} \right) \\ + \frac{1}{4} J \left(\sqrt{2 \left| F_{-\sqrt{P_1}, -\sqrt{P_2}} \left(\frac{1}{2} [J^{-1}(I_{vs}^{[e]}(j))]^2, \sigma_j^2 \right) \right|} \right). \quad (16)$$

• *V-to-c message update for JU*: The MI between the LLR-message from v_j to c_i and the sending symbol v_j is

$$I_{vc}^{[c]}(i, j) = J_R(\sigma_{v-to-c}). \quad (17)$$

$$J_R(\sigma_{v-to-c}) = I(S; \mathbf{W} = \mathbf{X} + \mathbf{Y}), \quad (18)$$

and

$$\sigma_{v-to-c}^2 = \sum_{u \in N(j), u \neq i} b_{uj} [J_A^{-1}(I_{cv}^{[c]}(u, j))]^2 \\ + (b_{ij} - 1) [J_A^{-1}(I_{cv}^{[c]}(i, j))]^2, \quad (19)$$

where $\mathbf{X}$ is a Gaussian random vector with distribution satisfying (10), $\mathbf{Y}$ denotes an initial message vector from (16), and $I_{vc}^{[c]}(i, j) = 0$ if $b_{ij} = 0$.

• *C-to-v message update for JU*: The MI between the LLR-message from c_i to v_j and the sending symbol v_j is given by

$$I_{cv}^{[c]}(i, j) = 1 - J_A(\sigma_{c-to-v}), \quad (20)$$

with

$$\sigma_{c-to-v}^2 = \sum_{u \in N(i), u \neq j} b_{iu} [J_A^{-1}(1 - I_{vc}^{[c]}(i, u))]^2 \\ + (b_{ij} - 1) [J_A^{-1}(1 - I_{vc}^{[c]}(i, j))]^2. \quad (21)$$

The sending symbol v_j is represented as a Gaussian random vector with parameter σ_{c-to-v} satisfying (10), and $I_{cv}^{[c]}(i, j) = 0$ if $b_{ij} = 0$.

• *Initial information for RU (message extraction)*: By employing (12) and (13), we extract the marginal probabilities of $v_j^{[2]}$ to derive

$$F_{+\sqrt{P_1}, +\sqrt{P_2}}(\mu, \sigma_j^2) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-z^2} \log \left(\frac{1 + e^{\sqrt{4\mu + \frac{8(\sqrt{P_1} + \sqrt{P_2})^2}{\sigma_j^2}} z + \mu + \frac{2(\sqrt{P_1} + \sqrt{P_2})^2}{\sigma_j^2}}}{1 + e^{-\sqrt{4\mu + \frac{8(\sqrt{P_1} - \sqrt{P_2})^2}{\sigma_j^2}} z - \mu - \frac{2(\sqrt{P_1} - \sqrt{P_2})^2}{\sigma_j^2}}} \right) dz - \mu - \frac{4\sqrt{P_1}\sqrt{P_2} + 2P_2}{\sigma_j^2} \quad (6)$$

$$F_{+\sqrt{P_1}, -\sqrt{P_2}}(\mu, \sigma_j^2) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-z^2} \log \left(\frac{1 + e^{\sqrt{4\mu + \frac{8(\sqrt{P_1} + \sqrt{P_2})^2}{\sigma_j^2}} z + \mu + \frac{2(P_1 - P_2)}{\sigma_j^2}}}{1 + e^{-\sqrt{4\mu + \frac{8(\sqrt{P_1} - \sqrt{P_2})^2}{\sigma_j^2}} z - \mu + \frac{2(\sqrt{P_1} - \sqrt{P_2})^2}{\sigma_j^2}}} \right) dz - \mu - \frac{4\sqrt{P_1}\sqrt{P_2} - 2P_2}{\sigma_j^2} \quad (7)$$

$$F_{-\sqrt{P_1}, +\sqrt{P_2}}(\mu, \sigma_j^2) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-z^2} \log \left(\frac{1 + e^{\sqrt{4\mu + \frac{8(\sqrt{P_1} + \sqrt{P_2})^2}{\sigma_j^2}} z + \mu + \frac{2(P_2 - P_1)}{\sigma_j^2}}}{1 + e^{-\sqrt{4\mu + \frac{8(\sqrt{P_2} - \sqrt{P_1})^2}{\sigma_j^2}} z - \mu - \frac{2(\sqrt{P_2} - \sqrt{P_1})^2}{\sigma_j^2}}} \right) dz - \mu - \frac{2P_2}{\sigma_j^2} \quad (8)$$

$$F_{-\sqrt{P_1}, -\sqrt{P_2}}(\mu, \sigma_j^2) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-z^2} \log \left(\frac{1 + e^{\sqrt{4\mu + \frac{8(\sqrt{P_1} + \sqrt{P_2})^2}{\sigma_j^2}} z + \mu - \frac{2(\sqrt{P_1} + \sqrt{P_2})^2}{\sigma_j^2}}}{1 + e^{-\sqrt{4\mu + \frac{8(\sqrt{P_1} - \sqrt{P_2})^2}{\sigma_j^2}} z - \mu - \frac{2(P_1 - P_2)}{\sigma_j^2}}} \right) dz - \mu + \frac{2P_2}{\sigma_j^2} \quad (9)$$

the prior LLR-message U for RU. Then, the initial MI of RU can be computed by

$$I_{ex}^{[e]}(j) = I(X; U). \quad (22)$$

The calculation of MI in RU is given in (14).

• *V-to-c message update for RU*: The message passed from v_j to c_i can be expressed as

$$I_{vc}^{[e]}(i, j) = J \left(\frac{\sqrt{\sum_{u \in N(j), u \neq i} b_{uj} [J^{-1}(I_{cv}^{[e]}(u, j))]^2}}{(b_{ij} - 1)[J^{-1}(I_{cv}^{[e]}(u, j))]^2 + [J^{-1}(I_{ex}^{[e]}(j))]^2} \right). \quad (23)$$

If $b_{ij} = 0$, $I_{vc}^{[e]}(i, j) = 0$.

• *C-to-v message update for RU*: The message passed from c_i to v_j is given by

$$I_{cv}^{[e]}(i, j) = 1 - J \left(\frac{\sqrt{\sum_{u \in N(i), u \neq j} b_{iu} [J^{-1}(1 - I_{vc}^{[e]}(i, u))]^2}}{(b_{ij} - 1)[J^{-1}(1 - I_{vc}^{[e]}(i, j))]^2} \right). \quad (24)$$

If $b_{ij} = 0$, $I_{cv}^{[e]}(i, j) = 0$.

• *Iterative termination conditions*: The convergence of JU and RU can be determined as follows.

1) *JU*: Given to the v-to-c message update for JU, the posteriori information is computed by

$$I_{app}^{[c]}(j) = J_R(\sigma_{app}). \quad (25)$$

with

$$\sigma_{app}^2 = \sum_{i=1}^M b_{ij} [J^{-1}(I_{cv}^{[c]}(i, j))]^2. \quad (26)$$

2) *RU*: The posteriori information for RU is computed by

$$I_{app}^{[e]}(j) = J \left(\sqrt{\sum_{i=1}^M b_{ij} [J^{-1}(I_{cv}^{[e]}(i, j))]^2 + [J^{-1}(I_{ex}^{[e]}(j))]^2} \right). \quad (27)$$

The system achieves convergence when $I_{app}^{[c]}(j) = 1$ and $I_{app}^{[e]}(j) = 1$, $\forall j$.

In this work, we consider the block fading channel. The proposed design can be extended to time-varying channels, i.e., multi-block fading channel and fast fading channel via block-wise equalization [18], [19]. Note that for the codeword with limited length spans a multiple fading sub-blocks, root-LDPC codes can be designed to achieve full diversity and perform close to the outage limit [20], which we leave for future work.

C. The Design Principles

Our target is to design base matrices $\mathbf{B}^{[c]} = (b_{i,j})_{M_1 \times N}$ and $\mathbf{B}^{[e]} = (b_{i,j})_{(M_2 - M_1) \times N}$ for JU and RU, respectively. The desirable base matrices should possess a low decoding threshold, which is the minimum channel quality required for reliable decoding [21]. Based on this criterion, the target rate pair is selected to achieve the best decoding threshold while maximizing the sum rate. In the rate-diverse system, the design principles for the protograph codes are as follows.

- 1) Determine the numbers of check nodes and variable nodes given the code rate.
- 2) To achieve a low decoding threshold, the protograph codes contain a certain portion of degree-2 variable nodes, and one variable node has a large degree to speed the converge. Note that we also consider degree-one variable node.
- 3) The degrees of the remaining variable nodes are set to no less than 3 to satisfy the linear minimum distance growth property.
- 4) Among all the base matrices that satisfy the constraints, select the one with the lowest decoding threshold.

IV. SIMULATION RESULTS

In this section, guided by the proposed RDJP-EXIT, we design and search for protograph-based LDPC codes. Similar to [6], in the base matrix, we set the maximum element to 3 to simplify the search except for one large-degree variable node.

We first consider the GMAC channel with $P_1 = 1.5$ and $P_2 = 0.9$. For two users with the rates of 3/5 and 2/5, after search, the base matrices for $\mathbf{H}_c$ and $\mathbf{H}_e$ are given by

$$B_{R=3/5}^{[c]} = \begin{bmatrix} 3 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 2 & 1 & 1 & 1 & 0 & 2 & 0 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 & 0 & 2 & 0 & 1 & 1 \\ 2 & 0 & 1 & 1 & 1 & 0 & 0 & 2 & 0 & 1 \end{bmatrix}, \quad (28)$$

$$B_{R=4/5}^{[e]} = \begin{bmatrix} 3 & 0 & 1 & 2 & 1 & 0 & 1 & 0 & 2 & 1 \\ 2 & 1 & 1 & 0 & 0 & 1 & 2 & 1 & 1 & 0 \end{bmatrix}. \quad (29)$$

To show the generality of the design, we consider one more GMAC channel with $P_1 = \sqrt{2}$ and $P_2 = 0.8$. The matrices for user rates $R_1 = 2/3$ and $R_2 = 1/2$ are designed as

$$B_{R=2/3}^{[c]} = \begin{bmatrix} 3 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 2 & 2 & 0 \\ 3 & 1 & 1 & 0 & 1 & 2 & 0 & 2 & 2 & 0 & 0 & 1 \\ 2 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 2 \end{bmatrix}, \quad (30)$$

$$B_{R=5/6}^{[e]} = \begin{bmatrix} 3 & 1 & 1 & 0 & 1 & 2 & 0 & 2 & 1 & 0 & 1 & 0 \\ 2 & 2 & 0 & 2 & 1 & 0 & 2 & 1 & 0 & 1 & 2 & 1 \end{bmatrix}. \quad (31)$$

We employ the progressive edge growth (PEG) algorithm [21] to lift the protograph for the desired blocklength. The number of channel decoding iterations and outer iterations between JUD and RUD are 50 and 3, respectively.

Fig. 3 shows the performance of the proposed matrices in a GMAC with $P_1 = 1.5$, $P_2 = 0.9$ and a length $L = 2100$. We select two representative LDPC codes, i.e., the EXIT-optimized irregular codes and AR4JA codes [5], [22], [11]. The degree distributions of the irregular codes are summarized in Table I. We observe that the proposed protograph codes outperform the AR4JA codes by approximately 0.8 dB and the irregular codes by 0.4 dB. It also achieves a performance gain of 0.15 dB over the joint protograph (JP) code in [6] since the latter is not specifically designed for RDJD. Then, we simulate the performance of matrices (30) and (31) with $P_1 = \sqrt{2}$ and $P_2 = 0.8$, and $L = 2700$. Fig. 4 shows that the proposed codes achieve performance gains of 1 dB and 0.4 dB over the AR4JA and irregular codes, respectively, at a BER of 10^{-4} .

The capacity region of a two-user GMAC is defined by [23]. Given a rate pair, we obtain the minimum SNR required for each code to achieve error-free transmission in the simulation as the code length $L \rightarrow \infty$ [23]. Then, we compute the capacity region for this SNR. Fig. 5 shows that the sum rate for the channel of $P_1 = 1.5$, $P_2 = 0.9$ is

TABLE I
IRREGULAR CODES FOR COMPARISON

	U_{user}	R	d_c	λ_2	λ_3	λ_6	λ_7	λ_8	λ_9	λ_{10}	λ_{20}	λ_{21}	λ_{23}
$P_1 = 1.5, P_2 = 0.9$	JU	3/5	10	0.1693	0.3711	0	0	0	0.3074	0	0.1119	0.0403	0
	RU	4/5	18	0.1296	0.5015	0	0.0090	0.3599	0	0	0	0	0
$P_1 = \sqrt{2}, P_2 = 0.8$	JU	2/3	13	0.1346	0.3758	0	0	0	0.2591	0.0426	0	0	0.1879
	RU	5/6	20	0.1435	0.5736	0.2829	0	0	0	0	0	0	0

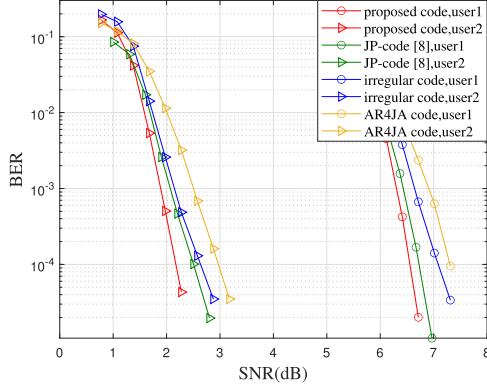


Fig. 3. Performance of the proposed protograph codes and other LDPC codes in a GMAC with $P_1 = 1.5, P_2 = 0.9$.

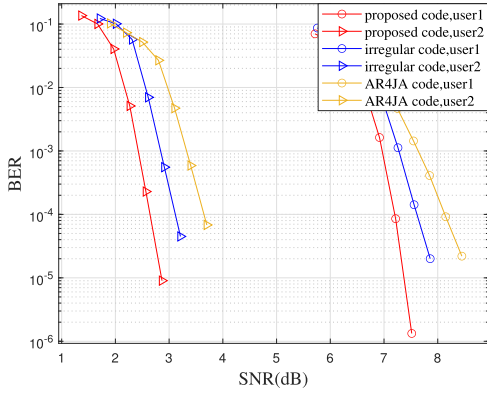


Fig. 4. Performance of the proposed protograph codes and other LDPC codes in a GMAC with $P_1 = \sqrt{2}, P_2 = 0.8$.

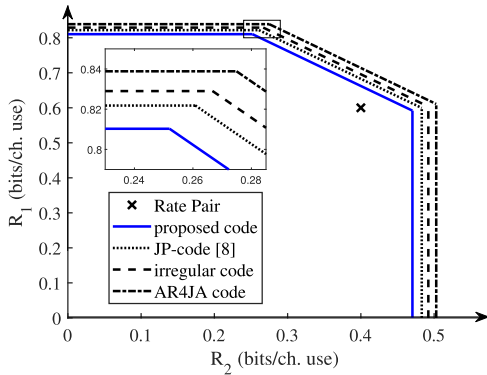


Fig. 5. Rate pairs and capacity regions with $P_1 = 1.5$ and $P_2 = 0.9$ for different schemes.

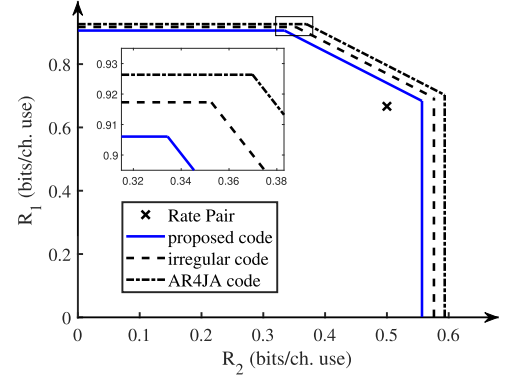


Fig. 6. Rate pairs and capacity regions with $P_1 = \sqrt{2}$ and $P_2 = 0.8$ for different schemes.

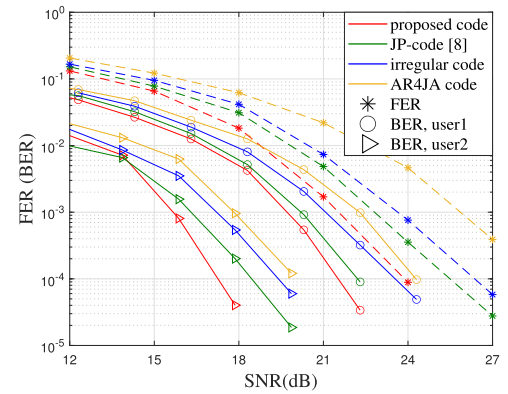


Fig. 7. Performance of the proposed codes and other LDPC codes over quasi-static Rayleigh fading channels with $E[|h_1|^2] = 1.5, E[|h_2|^2] = 0.9$.

1.062 bits/channel. The protograph codes (28) and (29) are only 0.062 bits/channel away from this capacity, outperforming the benchmark irregular code, which has a gap of 0.096 bits/channel. In addition, as shown in Fig. 6, the sum rate of the designed matrix set (30) and (31) is only 0.074 bits/channel away from the capacity region, compared to a gap of 0.102 bits/channel for the irregular code. The results demonstrate that the proposed protograph codes more closely approach the capacity limit than existing designs, achieving near-maximal sum rates.

We have also evaluated our proposed codes over quasi-static Rayleigh fading model, where (h_1, h_2) vary from one block to next one. The channel coherence time satisfies $T_c \geq N \times T_s$, where N is the block length and T_s is the symbol duration. This guarantees that each block experiences a constant fading realization. In addition to the BER metric used in Figs. 3 and 4, since random noise corrupts individual bits, here we also employ FER to characterize system performance, as a single deep fade may corrupt the entire

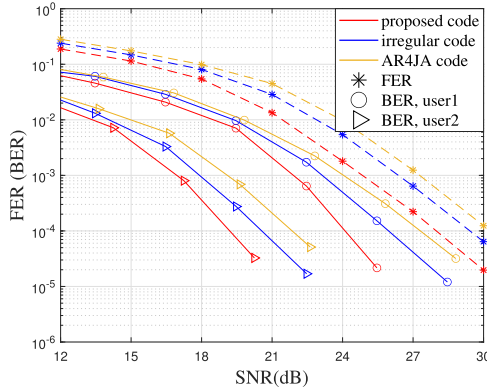


Fig. 8. Performance of the proposed codes and other LDPC codes over quasi-static Rayleigh fading channels with $E[|h_1|^2] = \sqrt{2}$, $E[|h_2|^2] = 0.8$.

codeword [24]. As shown in Figs. 7 and 8, our proposed codes in (28)–(29) and (30)–(31) significantly outperform their benchmark codes in the fading channels with $\{E[|h_1|^2], E[|h_2|^2]\} = \{1.5, 0.9\}$ and $\{E[|h_1|^2], E[|h_2|^2]\} = \{\sqrt{2}, 0.8\}$, respectively.

V. CONCLUSION

This paper proposes a design method for protograph-based LDPC codes for the rate-diverse GMAC. Considering the serial concatenated decoder at the receiver, we first analyze the message model of the system and propose the RDJP-EXIT analysis to evaluate the decoding behavior of the joint user and the residual user. Then, guided by the RDJP-EXIT analysis, the protograph matrices are designed for two users with low decoding thresholds for different channel qualities, respectively. Simulation results show that the proposed protograph codes can achieve better performance than the conventional AR4JA codes and irregular codes over different channels and approach the GMAC capacity region.

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